

Exploring Clickstream Data: A Guide Through Useful Business Insights for an E-commerce Site

Customer Behaviour • Review Influence • Purchase Activity • Customer Journey

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Programme / Context	Digital Marketing Analytics
Primary analytical tool	R programming

Executive Summary

This report analyses clickstream activity from an e-commerce website to understand customer behaviour, the influence of different product-review formats on purchasing activity, and the browsing journeys followed by users. The analysis combines clickstream feed data, product and review information, and registered-user demographics. The sample contains 16,123 unique users, five selling products, 16 URL types and activity recorded during the first 15 days of March 2012; approximately 51.6% of users are recorded as buyers.

The analytical workflow was implemented primarily in R and included data cleaning and transformation, session classification, user-level aggregation, statistical hypothesis testing, linear regression, page-frequency analysis and sequence-based customer-journey analysis. The R notebook also contains sequence-analysis, second-order Markov-chain modelling, hierarchical sequence clustering and sequence-level purchase-association modelling, extending the analysis beyond simple descriptive reporting.

The analysis indicates that customer reviews have the greatest visibility and a stronger observed relationship with purchase activity than video reviews or celebrity recommendations. Customer reviews generated approximately 68,074 sessions, compared with 41,974 for video reviews and 27,592 for celebrity recommendations. The review hypothesis tests did not support the proposition that celebrity reviews outperform customer reviews, while customer reviews performed significantly better than video reviews in the reported pairwise comparison.

Customer-journey analysis shows a marked difference between buyers and non-buyers: buyers visited approximately 1.65 distinct pages on average, while non-buyers visited approximately 8.52. This suggests that conversion is associated with a relatively short browsing path, whereas non-buyers tend to explore substantially more content before leaving. Gender-based browsing behaviour was comparatively similar, with approximately 10.38 pages visited by males and 10.73 by females.

The report recommends prioritising customer-generated reviews, improving the quality and placement of video and celebrity content, reducing unnecessary navigation between discovery and purchase, monitoring high-traffic and exit pages through clickstream analysis, and using controlled experimentation to test website interventions.

Key Management Takeaways

- Customer reviews are the most frequently encountered review format and show the strongest observed association with purchase activity.
- Celebrity endorsements should not be assumed to outperform customer-generated evidence simply because they are more aspirational.
- Buyer journeys are substantially shorter than non-buyer journeys, creating an opportunity to simplify the route to conversion.
- Clickstream data can support continuous website optimisation by revealing traffic concentration, navigation sequences and potential exit points.
- Future experimentation should test specific interventions—review placement, calls to action, product-page design and content sequencing—rather than relying on observational associations alone.

1. Introduction

In e-commerce, understanding how customers interact with a website is central to improving the user experience, increasing conversion and allocating marketing resources effectively. Clickstream data records the sequence of digital interactions made by users and therefore provides a behavioural view of the customer journey.

The project was designed around two connected business questions. First, it examined whether different sources of product reviews—customer reviews, video reviews and celebrity recommendations—differ in their relationship with purchasing activity. Second, it examined how users navigate through the website, including the pages they visit most often, typical browsing sequences, differences between buyers and non-buyers, and possible points of exit.

The overall objective was therefore not simply to describe website traffic, but to convert behavioural data into evidence that management could use to improve product presentation, website navigation, personalisation and conversion.

2. Problem Description and Business Objectives

The central business challenge is to understand what drives online purchasing activity and how the website can be structured to move customers more efficiently from product discovery to purchase.

2.1 Business Questions

1. Which type of product review receives the greatest exposure on the website?
2. Are celebrity recommendations more influential than customer reviews in determining conversions?
3. How do celebrity, video and customer reviews compare with one another in terms of observed purchasing impact?
4. What is the relationship between review exposure and a user's total number of purchases?
5. Which pages and navigation sequences characterise the customer journey?
6. How does browsing behaviour differ between buyers and non-buyers and between male and female users?
7. Where can the business intervene to improve customer experience and conversion?

2.2 Business Significance

The answers have direct commercial relevance. Review placement can influence how product credibility is communicated; understanding high-traffic pages can inform website design; and understanding the routes taken by buyers and non-buyers can help the business identify opportunities to simplify navigation. The original presentation framed these outputs around personalisation, targeting and conversion optimisation.

3. Data and Dataset Overview

Three principal datasets were used: clickstream feed data, products and reviews data, and registered-user data. The clickstream feed contains user/session identifiers, URLs, timestamps and a binary purchase indicator. The product and review information maps website locations to review or product categories, while the registered-user data supplies demographic information such as age and gender.

Data element	Description	Business use
Clickstream feed	User/session, URL, timestamp and purchase indicator	Reconstruct journeys and purchasing behaviour
Products & reviews	URL/review identifiers and review categories	Classify pages and compare review exposure
Registered users	User demographics including gender	Segment browsing behaviour

3.1 Dataset Profile

Metric	Observed value
Unique users	16,123
Selling products	5
URL types	16

Metric	Observed value
Website landing-page categories	5
Review types	3
Recorded buyers	~51.6% of users
Observation period	First 15 days of March 2012

3.2 Review Categories

The website contains three principal review formats. Customer reviews provide written product feedback; video reviews provide recorded reactions and demonstrations; and celebrity recommendations provide influencer or endorsement-based credibility. The R notebook maps review identifiers into these analytical categories before aggregation.

Code Extract 1. Mapping raw review/page identifiers to business categories

```
clickstream <- clickstream %>%
  mutate(
    category = case_when(
      review_id == 1000 ~ "home page",
      review_id %in% c(1001, 1002, 1003) ~ "video review",
      review_id %in% c(2001, 2002) ~ "celebrity recommendation",
      review_id %in% c(3001, 3002, 3003, 3004, 3005) ~ "customer review",
      review_id %in% c(4001, 4002, 4003, 4004, 4005) ~ "product",
      TRUE ~ NA_character_
    )
  )
```

4. Analytical Methodology

The analytical process followed four broad stages: data preprocessing, statistical analysis, customer-journey analysis and business interpretation. This follows the structure presented in the original project materials.

4.1 Data Preparation

- Imported clickstream, product and registered-user datasets into R.
- Joined registered-user attributes to clickstream records using the user/session identifier.
- Classified each session/page interaction into home page, video review, celebrity recommendation, customer review or product.
- Grouped observations by review category, review identifier and user/session.
- Created user-level review exposure measures and a total-purchases measure.
- Ordered clickstream records by user and timestamp to enable journey reconstruction.

Code Extract 2. Joining datasets and constructing user-level review exposure variables

```
merged_data <- clickstream %>%
  left_join(regusers, by = "user_session_id")
```

```

review_exposure <- merged_data %>%
  group_by(user_session_id) %>%
  summarise(
    customer_review = sum(category == "customer review", na.rm = TRUE),
    video_review = sum(category == "video review", na.rm = TRUE),
    celebrity_recommendation = sum(
      category == "celebrity recommendation",
      na.rm = TRUE
    ),
    total_purchases = sum(is_purchased., na.rm = TRUE),
    purchased = as.integer(any(is_purchased. == 1, na.rm = TRUE)),
    .groups = "drop"
  ) %>%
  mutate(
    customer_review_exposed = as.integer(customer_review > 0),
    video_review_exposed = as.integer(video_review > 0),
    celebrity_review_exposed = as.integer(
      celebrity_recommendation > 0
    )
  )

```

4.2 Statistical Analysis

Hypothesis tests were used to compare the observed conversion impact of different review types. The project used right-tailed standard-normal tests for the stated pairwise comparisons at a 5% significance level. A linear regression model was also used to examine the relationship between total purchases and review exposure.

Code Extract 3. Regression model used to relate purchase activity to review exposure

```

purchase_model <- lm(
  total_purchases ~
    customer_review +
    video_review:celebrity_recommendation,
  data = review_exposure
)

summary(purchase_model)

```

The original presentation reports coefficients of 0.1814 for customer-review exposure and 0.003761 for the combined video/celebrity term. The project interprets the larger customer-review coefficient as evidence of a stronger observed relationship with purchase activity.

4.3 Customer-Journey Analysis

Customer journeys were reconstructed by sorting records by user and timestamp and then analysing the sequence of URLs visited. The analysis included page-frequency counts, randomly selected user paths and sequence-based modelling. The R notebook also contains a second-order Markov-chain model, hierarchical clustering of clickstream sequences and sequence-level purchase-association modelling.

Code Extract 4. Constructing page sequences and fitting a second-order Markov chain

```
merged_data <- merged_data %>%
  arrange(user_session_id, timestamp)

page_counts <- merged_data %>%
  count(url, sort = TRUE)

head(page_counts, 10)
```

Code Extract 5. Reconstructing individual customer paths

```
set.seed(123)

random_user_ids <- sample(
  unique(merged_data$user_session_id),
  size = 5
)

sampled_journeys <- merged_data %>%
  filter(user_session_id %in% random_user_ids) %>%
  arrange(user_session_id, timestamp)

sequence_data <- sampled_journeys %>%
  group_by(user_session_id) %>%
  summarise(
    page_sequence = paste(url, collapse = " > "),
    exit_page = last(url),
    .groups = "drop"
  )

sequence_data
```

5. Analysis 1 — Influence of Product Reviews on Purchasing

The first analytical question was whether different forms of product reviews have different relationships with customer purchasing behaviour. The analysis considered the frequency of exposure to customer reviews, video reviews and celebrity recommendations, followed by hypothesis testing and regression.

5.1 Review Exposure

Review type	Approx. sessions	Interpretation
Customer review	68,074	Highest visibility
Video review	41,974	Second-highest visibility
Celebrity recommendation	27,592	Lowest visibility

Customer reviews were encountered most frequently. The original presentation also showed that review types were not distributed uniformly across all products: products 1 and 2 had a more balanced mix, product 3 had no celebrity recommendations, and products 4 and 5 showed customer

reviews only. This uneven product-level availability is important when interpreting review-type comparisons.

5.2 Celebrity Reviews versus Customer Reviews

The stated hypothesis tested whether celebrity reviews were more influential than customer reviews. The reported right-tailed test produced a z-score of -145.03 and a p-value approximately equal to 1. At the 5% significance level, the null hypothesis was therefore not rejected. The project concludes that celebrity recommendations were not shown to outperform customer reviews in determining conversions.

5.3 Pairwise Comparison with Video Reviews

The pairwise tests provide a more differentiated picture. Celebrity reviews did not outperform video reviews, with a reported z-score of -58.63 and p-value approximately equal to 1. In contrast, customer reviews performed better than video reviews, with a reported z-score of 88.71 and p-value below 0.05.

Comparison	Test statistic	Reported p-value	Project conclusion
Celebrity vs customer	-145.03	≈ 1	No evidence celebrity is more influential
Celebrity vs video	-58.63	≈ 1	Celebrity does not outperform video
Customer vs video	88.71	< 0.05	Customer reviews perform better

5.4 Regression Analysis of Purchase Activity

To quantify the relationship between review exposure and purchase activity, the project modelled total purchases at user level using customer-review exposure and the interaction between video and celebrity-review exposure. The reported coefficients were 0.1814 for customer reviews and 0.003761 for the combined video/celebrity term.

The practical interpretation in the project is that customer-generated feedback has a stronger observed relationship with purchase activity than the combined video/celebrity exposure measure. This supports prioritising authentic customer feedback while improving the effectiveness of other review formats.

Analytical caution: this regression is observational. The coefficient should be interpreted as an association with purchase activity, not as proof that exposure to a particular review type causes additional purchases.

6. Analysis 2 — Determining the Customer Journey

The second business question focused on how customers move through the website. Rather than examining pages in isolation, the analysis reconstructed user-level sequences so that the business could understand common paths, high-traffic pages, and differences in navigation between customer groups.

6.1 Most Visited Pages

The homepage is the most frequently visited location, followed by several product pages. Products 3, 1 and 5 were identified as particularly popular among users in the presentation. These traffic concentrations provide a starting point for deciding where merchandising, review placement and conversion calls-to-action may have the greatest reach.

6.2 Example Customer Path

The project reconstructs individual paths by ordering events by timestamp. One illustrated journey begins at the homepage, moves through a product page and customer review, continues to another product and video-review page, and then exits. Such sequences provide a practical way to identify where customers move from discovery into evaluation and where browsing ends.

6.3 Buyer versus Non-Buyer Behaviour

Customer segment	Average distinct pages visited	Business interpretation
Buyers	~1.65	Shorter path to conversion
Non-buyers	~8.52	Substantially more exploration before exit
Male users	~10.38	Similar browsing intensity
Female users	~10.73	Similar browsing intensity

The difference between buyers and non-buyers is one of the most commercially useful findings in the project. Buyers appear to reach a purchase with relatively little navigation, while non-buyers interact with substantially more pages. This may indicate that a streamlined route is associated with successful purchase behaviour, although the analysis does not establish that reducing page visits would itself cause higher conversion.

6.4 Sequence and Markov Analysis

The R notebook extends the journey analysis through sequence methods. URLs are converted into user-level sequence objects and their frequencies are examined. A second-order Markov chain is then fitted to the clickstream list, allowing the analysis to model the probability of moving to the next page conditional on the preceding browsing history. Hierarchical clustering and sequence-level purchase-association modelling are also included in the notebook.

This is an important analytical extension because it moves the project from static page counts towards modelling the structure of customer journeys. In a production setting, the same approach could be developed further to identify journey archetypes, detect high-friction paths and estimate the likelihood of conversion from different stages of a session.

7. Business Interpretation

7.1 What the Evidence Suggests

- Authentic customer feedback appears more commercially relevant than celebrity endorsements in the observed sample.
- The website's review inventory is uneven across products, so review strategy should be considered at product level rather than only as an overall website average.
- Customers who purchase tend to follow a much shorter path than those who do not purchase, suggesting that unnecessary navigation may be a potential friction point.
- The customer journey should be treated as a sequence rather than a collection of independent page visits.
- The strongest opportunities lie at the intersection of content credibility and journey simplicity: provide the evidence customers need while making the route to purchase easy.

7.2 Implications for Personalisation and Targeting

The observed differences in browsing behaviour create a foundation for personalisation. For example, a user who repeatedly moves between product pages and review pages could be shown more relevant product evidence or a clearer purchase call-to-action. Similarly, frequently visited product pages can be prioritised for stronger review placement and experimentation.

The current project does not build a production propensity model or recommendation engine. However, the clickstream structure provides the behavioural variables needed for a future model—for example, page depth, review exposure, product interactions, journey sequence, recency and session-level engagement.

8. Recommendations

Prioritise customer-generated reviews: Increase the visibility and quality of customer feedback on high-traffic product pages because customer reviews show the strongest observed relationship with purchase activity in the analysis.

Improve video and celebrity content rather than simply increasing volume: The results do not support assuming that celebrity endorsements are superior. Test whether stronger video quality, credibility, placement and relevance can improve their contribution.

Shorten the path from discovery to purchase: Review product pages, navigation and calls-to-action to identify unnecessary steps. Use the buyer journey as the benchmark for a more efficient conversion path.

Use sequence analytics to identify friction points: Monitor common page-to-page transitions, high-frequency exit pages and journey archetypes. The existing Markov and sequence-analysis framework provides a starting point for this monitoring.

Introduce controlled A/B testing: Test review placement, CTA wording, page layout and content sequencing. The original project recommends experimentation; future tests should use larger samples and random assignment wherever possible.

Build customer-level propensity features: Use page depth, review exposure, product interactions and sequence characteristics to develop a purchase-propensity model in Python/R. Evaluate it using an appropriate train/test framework and business metrics.

Segment by product and customer behaviour: Because review availability differs substantially by product, optimise content strategy at product level and consider behavioural segments rather than relying only on demographic segmentation.

9. Limitations and Future Scope

- The dataset covers only the first 15 days of March 2012, so the observed patterns may not represent longer-term behaviour.
- The original analysis uses observational clickstream data; statistical association should not be interpreted as causal impact.
- The review types are not evenly available across products, which may introduce product-level confounding into comparisons.
- The original project used a linear regression specification for purchase activity. A future analysis could consider a count model or a classification/propensity model depending on the precise target definition.
- The customer-journey analysis includes random examples of individual paths. A stronger production analysis would quantify journey clusters across the full population and validate them out-of-sample.
- The project identifies that buyers have shorter browsing paths, but this does not establish that shorter journeys cause conversion. Customers may differ in intent before entering the site.
- Future work could combine clickstream behaviour with product attributes, customer demographics and time-based features to develop customer-level propensity and recommendation models.
- A/B testing should use larger samples and a clearly defined primary conversion metric to increase statistical power and decision confidence.

10. Conclusion

This project demonstrates how clickstream data can be transformed into actionable business intelligence by combining data preparation, statistical analysis, regression and customer-journey modelling. The analysis finds that customer reviews are the most visible review format and show a stronger observed relationship with purchase activity than the other review formats examined. It also identifies a substantial behavioural difference between buyers and non-buyers: buyers follow a much shorter browsing path.

The commercial implication is not simply to add more reviews or remove pages. Rather, the business should use behavioural evidence to determine which information customers need, where they need it, and how quickly they can move from evaluation to purchase. Customer-generated evidence should be prioritised, other review formats should be improved and tested, and journey-level clickstream monitoring should be incorporated into ongoing website optimisation.

The project also establishes a strong analytical foundation for more advanced customer Data Science. The existing sequence and Markov-chain work could be extended into journey segmentation and propensity modelling, while controlled experimentation could move the analysis from observational insight towards causal decision-making.

References and Project Sources

8. Project R Notebook — [r code dma ind.nb.html](#). Source code and analytical outputs used throughout this report.
9. Project Presentation — DMA Individual PPT. Presentation of business problem, methodology, statistical results and customer-journey findings.
10. Project Presentation Script — [Script_DMA.pdf](#). Detailed narrative explaining the business interpretation of each analytical stage.
11. Digital Marketing Analytics Group Sample — [WA_A3_Part_A_Group5.pdf](#). Used as the structural reference for the business-report format, including executive summary, analytical sections, recommendations and conclusion.

Appendix A — Selected R Code Used in the Analysis

The following extracts are reproduced from the application-ready R Markdown analysis and integrated here to document the analytical workflow. They are selected rather than a full code dump so that the report remains a business document rather than a programming notebook.

Appendix Code A1. Analytical packages

```
library(dplyr)
library(tidyr)
library(TraMineR)
library(clickstream)
```

Appendix Code A2. Dataset import

```
data_dir <- params$data_dir

clickstream <- read.delim(
  file.path(data_dir, "clickstream-feed-generated.tsv"),
  sep = "\t",
  stringsAsFactors = FALSE
)

products <- read.delim(
  file.path(data_dir, "products (2).tsv"),
  sep = "\t",
  stringsAsFactors = FALSE
)

regusers <- read.delim(
  file.path(data_dir, "regusers (1).tsv"),
  sep = "\t",
```

```
stringsAsFactors = FALSE
)

names(regusers)[1] <- "user_session_id"
```

Appendix Code A3. User-level purchase model

```
purchase_model <- lm(
  total_purchases ~
    customer_review +
    video_review:celebrity_recommendation,
  data = review_exposure
)

summary(purchase_model)
```

Appendix Code A4. Page-frequency analysis

```
merged_data <- merged_data %>%
  arrange(user_session_id, timestamp)

page_counts <- merged_data %>%
  count(url, sort = TRUE)

head(page_counts, 10)
```

Appendix Code A5. Sequence and Markov-chain analysis

```
user_paths <- merged_data %>%
  group_by(user_session_id) %>%
  arrange(timestamp, .by_group = TRUE) %>%
  summarise(
    path = list(url),
    .groups = "drop"
  )

max_path_length <- max(lengths(user_paths$path))

sequence_matrix <- do.call(
  rbind,
  lapply(
    user_paths$path,
    function(x) {
      c(
        x,
        rep(NA_character_, max_path_length - length(x))
      )
    }
  )
)

sequence_matrix <- as.data.frame(sequence_matrix)
```

```

clickstream_seq <- seqdef(
  sequence_matrix,
  right = "DEL"
)

markov_lines <- user_paths %>%
  mutate(
    clickstream = paste(
      user_session_id,
      paste(path, collapse = ","),
      sep = ","
    )
  ) %>%
  pull(clickstream)

markov_file <- tempfile(fileext = ".csv")

writeLines(
  c("user_session_id", markov_lines),
  markov_file
)

cls <- readClickstreams(
  markov_file,
  header = TRUE
)

markov_model <- fitMarkovChain(
  clickstreamList = cls,
  order = 2,
  control = list(optimizer = "quadratic")
)

set.seed(123)
cluster_ids <- sample(
  seq_len(nrow(clickstream_seq)),
  size = min(1000, nrow(clickstream_seq))
)

cluster_seq <- clickstream_seq[cluster_ids, ]

clickstream_dist <- seqdist(
  cluster_seq,
  method = "OM"
)

clickstream_hclust <- hclust(
  as.dist(clickstream_dist),
  method = "ward.D2"
)

plot(clickstream_hclust)

```

